

Kushal R Gowda

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SUMMARY

Computer Science undergraduate (CGPA: 8.54) specializing in AI/ML systems and backend engineering, with hands-on experience building Retrieval-Augmented Generation (RAG) pipelines, multi-agent LLM orchestration systems, and low-latency backend infrastructure, plus industry experience in agentic and generative AI platform development. Proficient in Python, PyTorch, LangGraph, and FastAPI, with applied experience in Natural Language Processing (NLP), Machine Learning (ML), real-time data streaming (Kafka, Spark), and vector database architecture (Qdrant). Seeking a 12-month Co-op/Internship in AI/ML Engineering or Backend Development.

EDUCATION

Malnad College of Engineering (MCE)

Bachelor of Engineering (B.E.), Computer Science and Engineering — CGPA: 8.54

Hassan, Karnataka

Expected May 2027

- Eligible for 12-Month Co-op / Internship — available immediately

EXPERIENCE

AI Platform Development Engineer Intern — Webveer

- Contributed to the development of agentic AI and generative AI capabilities within the company's AI platform.
- Collaborated with the engineering team on building and refining LLM-driven agentic workflows and generative AI features.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript/TypeScript, SQL

AI/ML & NLP: Machine Learning (ML), Deep Learning, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), MLOps, Multi-Agent Systems, Prompt Engineering, LangGraph, PyTorch, ONNX Runtime, SentenceTransformers

LLM Integration: Google Gemini API (1.5 Pro/Flash), Groq API, Llama 3.1, Semantic Vector Search

Backend & Systems: FastAPI, Node.js, Express.js, REST API Design, Asynchronous Programming, WebSockets, Event-Driven Architecture, Distributed Systems, React.js

Data Engineering: Apache Kafka (KRaft), Apache Spark (PySpark), PostgreSQL, Qdrant (Vector DB), SQLite

Tools & DevOps: Docker, Git, CI/CD Pipeline Management, System Design, Data Structures & Algorithms (DSA), Software Testing (Pytest, Unit Testing)

Certifications: AWS Certified Cloud Practitioner

TECHNICAL PROJECTS

UPI Mule-Ring Sentinel | XGBoost, NetworkX, FastAPI, Pandas, Pytest

2026

- Engineered an 18-feature graph pipeline (cycle participation, PageRank, Louvain closure) over a 3,000-account synthetic UPI network, reaching 0.974 ROC-AUC at 0.87 recall and 0.43 precision on a ring-disjoint held-out test split.
- Chose the threshold on validation by minimizing a rupee cost matrix (₹2,00,000 per miss vs ₹15,000 per false alert) rather than F1: ₹4.72M cost against ₹12.8M for the 0.98-precision alternative, within 4.9% of hindsight-optimal.
- Constrained the service to a defense-only action ceiling through three independent rails and an import-time enum sweep, and published a 26-finding audit of my own codebase alongside a 375-case test suite.

Omni-Agent Trading Nexus | LangGraph, Gemini API, FastAPI, WebSockets, React

2026

- Architected a 4-agent financial swarm in LangGraph, parallelizing sentiment analysis and technical parsing to reduce trade signal generation latency by 40%.
- Built a resilient LLM orchestration pipeline with Gemini API model cascading and exponential backoff, maintaining 99.8% uptime against HTTP 429 throttling.

Aegis-Edge | PyTorch, ONNX Runtime, C++, Kafka, PySpark

2025 – 2026

- Ingested high-frequency 50Hz CNC vibration telemetry via a native Kafka (KRaft) broker, streaming real-time micro-batches in PySpark to trigger automated kill-switches at >85% risk thresholds.
- Compiled PyTorch 1D-CNN models into a C++ ONNX Runtime inference pipeline, bypassing Python serialization overhead to achieve sub-5ms edge latency.
- Implemented custom C++ memory pooling and INT8 quantization, reducing peak RAM consumption by 25% and boosting execution speed to 45+ FPS on edge hardware.

VIGIL-SOAR | LangGraph, Groq API (Llama 3.1), Pydantic, Python

2026

- Engineered an event-driven Security Orchestration (SOAR) pipeline utilizing LangGraph to process and triage SIEM network logs in real time.
- Distributed cognitive load across 3 specialized Groq (Llama 3.1) agents, accelerating Active Directory forensics and threat response times by 65%.

Zero-Trust RAG Engine | LangGraph, Qdrant, Docker, FastAPI, Gemini API

2026

- Enforced RBAC at the vector search layer via Qdrant, using strict metadata pre-filtering to isolate tenant data and eliminate LLM context leakage.
- Orchestrated a deterministic LangGraph state machine to short-circuit unauthorized queries, bypassing the Gemini API to prevent hallucinated disclosures.